

Question:

A 16-year-old girl is brought to the physician because menarche has not yet occurred. She has no history of serious illness and takes no medications. She is 162 cm (5 ft 3 in) tall and weighs 80 kg (176 lb); BMI is 31.2 kg/m². Breast and pubic hair development is Tanner stage 4. She also has oily skin, acne, and hyperpigmentation of the intertriginous areas of her neck and axillae. The remainder of the examination, including pelvic examination, shows no abnormalities. Which of the following is the most likely explanation for this patient's amenorrhea? Answer Choices: A. XO chromosomal abnormality B. Müllerian agenesis C. Elevated LH:FSH ratio D. Elevated β -hCG levels

Reasoning + Answer:

<think>The patient's BMI of 31.2 kg/m² indicates obesity, which can be associated with polycystic ovary syndrome (PCOS), a condition that can cause elevated LH:FSH ratio, acne, and hyperpigmentation, and is also often accompanied by amenorrhea due to irregular or absent ovulation, which fits the description of the patient's symptoms including the oily skin, acne, and hyperpigmentation of intertriginous areas as well as the lack of menarche and other physical signs of PCOS such as hirsutism and irregular menstrual cycles.</think><answer>C. Elevated LH:FSH ratio</answer>

